# Two measurable laws for dynamic optimization: when exploration pays and when sharing searches

Nam H. Le

*Department of Computer Science and Vermont Complex Systems Research Institute, University of Vermont, Burlington, VT, USA*

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**Abstract**

Optimizers for changing objectives rely on two devices: a budget of global exploration, and an information-sharing channel that spreads good solutions through the population. How much of each a problem deserves is set today by heuristics (hypermutation triggers, random immigrants, diversity repair), each tuned per benchmark. We derive two laws that replace the tuning with measurement. First, an exploration budget pays if and only if the rate of relevant environmental change exceeds a threshold $\psi^* = c'/(P'\Delta)$ built from three quantities an optimizer can estimate during its own run: the marginal cost of a diverted evaluation, the marginal probability that exploration recovers the optimum, and the value of recovery. Second, a sharing channel searches, rather than merely transmits, only while the pool it samples from retains variance commensurate with the displacement it must cover; below that it is a conduit, and no channel can create a reason to explore in a static world. On the moving-peaks benchmark the measured threshold reproduces the empirical sign of exploration's benefit across the change-rate $\times$ severity plane, and reveals that over most of that plane exploration is priced not as a mean improvement but as tail insurance: it leaves the median cost visible and removes the catastrophic stagnations. The same structure recurs on a dynamic knapsack measured against exact dynamic-programming optima, and the conduit bound is attained exactly once a recombination channel's pool has converged.

*Keywords:* dynamic optimization, exploration–exploitation, particle swarm optimization, parameter control, diversity, risk

*Email address:* nam.le@uvm.edu (Nam H. Le)

## 1. Introduction

A population-based optimizer facing a changing objective must answer two questions continuously. How many evaluations should it divert from exploiting its incumbent solution to exploring for displaced optima? And can it trust its information-sharing machinery, from the global best of a swarm to the parent pool of a crossover or the model of an estimation-of-distribution algorithm, to do any of that exploring on its behalf?

The field's answers are heuristics. Triggered hypermutation raises the mutation rate when a change is detected [1]; random immigrants inject fresh solutions at a fixed rate [2]; multi-swarm methods patrol for new peaks [3]; self-adaptive schemes let the run tune its own parameters [4, 5]. Each embeds an implicit judgement about how much exploration the problem's dynamics deserve, and each leaves that judgement to the practitioner or to a meta-search. Surveys of evolutionary dynamic optimization have noted for two decades that the field quantifies its *algorithms* far more precisely than its *reasons* [6, 7].

This paper supplies two of the missing reasons as measurable laws. Both are statements about where the benefit of a mechanism comes from, and both reduce, at the boundary, to familiar practice. The first law locates the entire benefit of exploration in the nonstationarity of the problem and prices its cost in diverted evaluations, yielding an explicit threshold on the change rate below which exploration is pure overhead; the static limit of the law is the standard advice to anneal exploration away. The second law concerns the sharing channel: sharing lowers the cost side of the first law, it is *permissive*, but cannot manufacture benefit, and the channel's own capacity to search is bounded by the variance of the pool it samples, a quantity that collapses exactly when premature convergence sets in.

Three things distinguish this from the surrounding literature. The quantities in both laws ($\psi$, $\Delta$, $P'$, $c'$, pool variance $V_d$) are defined operationally and estimated online, from the run itself, with no oracle access to the benchmark. The laws make sign predictions (exploration helps here, is overhead there) that are tested cell by cell against a change-rate $\times$ severity phase plane rather than against a single tuned setting. And the failure of the mean as a summary is part of the result: over most of the plane the median run pays a small, visible cost for exploration while the tail of catastrophic stagnations

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is removed, so exploration prices as insurance in the sense of conditional value-at-risk [8], not as expected improvement.

Section 2 states the laws; Section 3 describes the testbed and the online estimators; Section 4 tests the threshold, the insurance reading, a minimal controller, the conduit criterion in both its refuted and exact forms, and the combinatorial replication; Sections 5–6 position and qualify the results.

## 2. Two laws

### *2.1. Setting*

Let a population optimizer spend a fixed budget of $E$ evaluations per generation on an objective $f_t$ that changes at rate $\psi$ (relevant changes per evaluation). When a change occurs, the incumbent's quality drops by $\Delta$ on average (the *value of recovery*), and a per-generation exploration budget of $m$ evaluations recovers the optimum with probability $P(m)$, increasing and concave, at an opportunity cost $c(m)$: the performance the diverted evaluations would have produced in their best alternative use.

### *2.2. Law 1: the exploration threshold*

**Law 1 (Threshold).** *The expected per-generation gain of budget $m$ is*

$$G(m) \;=\; \psi\, P(m)\, \Delta \;-\; c(m), \tag{1}$$

*so exploration pays at all if and only if*

$$\psi \;>\; \psi^{*} \;:=\; \frac{c'(0)}{P'(0)\,\Delta}, \tag{2}$$

*and the optimal budget solves $\psi P'(m^{*})\Delta = c'(m^{*})$.*

Every quantity on the right of (2) is estimable during the run: $c'$ from the known evaluation accounting, $\Delta$ from the measured drop at each detected change, $P'$ from the bookkeeping of exploration successes, the same bookkeeping that underlies the classical 1/5th success rule [12]. At $\psi = 0$ the gain is $-c(m) \le 0$: in a static world exploration beyond convergence never pays, which is the annealing advice in its familiar form.

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*2.3. Law 2: sharing is permissive, and searches only with variance*

**Law 2 (Permissive channel).** *A sharing channel of strength $s$ (archives, copied solutions, model sampling) acts on (1) only through $P(\cdot)$ and $c(\cdot)$: it lowers $\psi^*$. It cannot change the sign of $G$ at $\psi = 0$.*

**Law 3 (Conduit criterion).** *A* sampling *channel, one whose proposals are drawn from its demonstrator pool, can only propose solutions inside the span of that pool. For a recombination channel on binary strings the span is the subcube spanned by the pool's support, so the attainable value after a change is bounded by the optimum restricted to that support, a quantity computable from the pool at the moment of change; for a Gaussian-resampling channel the span is set by the pool variance $V_d$, and relocating an optimum displaced by $\delta$ requires $V_d \gtrsim \kappa\,\delta^2$. Outside its span the channel transmits but cannot discover. The criterion applies to sampling channels only; a channel with its own dynamics (e.g. particle velocities) is not bounded by pool variance, and Section 4.6 shows this distinction empirically in both directions.*

**Corollary 1 (Insurance).** *Near and below $\psi^*$, $G(m) < 0$ in the median run while exploration still truncates the upper tail of the error distribution caused by rare stagnation events; the correct price of exploration there is a reduction of conditional value-at-risk, not of expected error.*

The corollary is stated informally here and established empirically in Section 4; its mechanism is that recovery failures are absorbing (a stranded population stays stranded) so their cost grows with the time to the next rescue, fattening the tail.

## 3. Experimental setting

*Benchmark.* Moving Peaks [9], $d = 5$, ten conic peaks (heights $[30, 70]$, widths $[1, 12]$, coordinates $[0, 100]^d$), change every $\tau$ evaluations: every peak drifts by a shift vector of length $s$=1 and its height moves by $\mathcal{N}(0, h_{\text{sev}})$. The height severity $h_{\text{sev}}$ controls how often the identity of the global peak changes (reordering); a single-peak row anchors the easy-static limit. $\psi = 1/\tau$.

*Base optimizer.* A constriction PSO [10, 11] ($\chi = 0.7298$, $c_1$=$c_2$=2.05, 20 particles) with a small local refinement budget ($g$=5 Gaussian (1+1) steps around the incumbent per generation) and a sentinel: the incumbent position

is re-evaluated at the start of every generation, anchored to the value it had at the previous generation start, so that a change is detected within one generation (98–99% of changes detected; the anchoring matters, a naïve re-evaluation misses ∼40% by comparing against values already contaminated by post-change evaluations). On detection, all personal-best memories are re-evaluated.

*The knob.* The exploration budget under test is $m$ uniform *scout* evaluations per generation; a scout replaces the worst particle when it beats that particle's personal best. Local refinement is part of the base, not of the knob: refinement around the incumbent is exploitation and improves tracking at every change rate, so it carries no threshold. The threshold concerns *global* exploration only, a distinction Law 1 forces and the experiments confirm.

*Estimators.* $\hat{\psi}$: detected changes per evaluation. $\hat{\Delta}$: mean drop of the sentinel value at detection. $\hat{P}'$: fraction of scout evaluations that improved on the incumbent. $\hat{c}'$: read from the sub-threshold plateau of the median benefit (below). All are causal; nothing reads the benchmark's internals.

## 4. Results

### *4.1. What tracking looks like*

Before any differences, the absolute picture. Figure 1 shows the error of the best solution found since the last change, averaged over seeds, for two representative cells. Each environmental change spikes the error to the height of the new gap; what distinguishes policies is the floor they recover to and how fast they reach it. Without exploration the swarm converges once, to whatever peak it first finds, and then tracks only that peak: its floor sits near 20 error units and never improves across periods. The three exploring policies recover to floors of 12–16 within a few hundred evaluations of each change and keep improving across the run as scouts relocate better peaks. On the knapsack, where the optimum is known exactly, the corresponding absolute statement is that the base GA holds 0.4–4% below the DP optimum depending on the regime, and the always-on budget roughly halves that deficit (Appendix Appendix A). All absolute errors, per cell and policy, with paired tests, are tabulated in Appendix Appendix A; the remainder of this section analyses the differences those tables summarise.

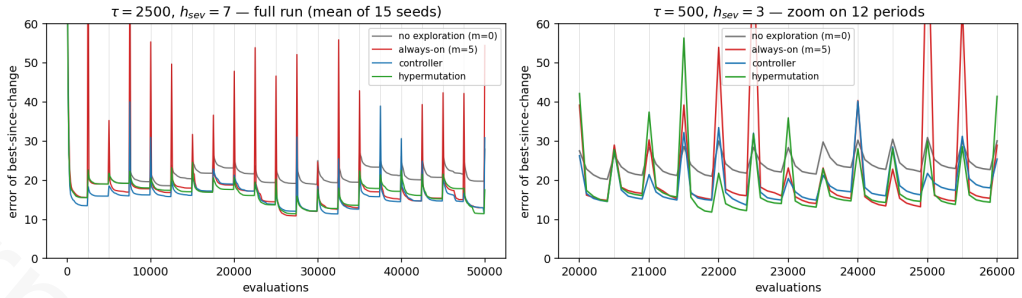



Figure 1: Error of the best solution found since the last change (mean of 15 seeds; vertical lines mark changes). Left: full run at $\tau$=2500, $h_{\text{sev}}$=7. Right: twelve change periods at $\tau$=500, $h_{\text{sev}}$=3. The no-exploration policy (grey) never improves its floor; the exploring policies re-converge within a few hundred evaluations of each change. Post-change spike heights are dominated by which evaluation lands first after the reset and carry no comparative meaning.

### *4.2. The sign of exploration across the phase plane*

We ran the base optimizer with and without the scout budget ($m \in \{0, 5\}$, paired seeds) over a $6 \times 5$ phase plane: change periods $\tau \in \{500, \ldots, 25000\}$ evaluations against five environment rows (a single-peak anchor, and ten peaks with height severity $h_{\text{sev}} \in \{0, 1, 3, 7\}$), 30 seeds per cell, 50,000 evaluations per run, 1,800 runs in all. Figure 2 (left) shows the median paired benefit.

Three features carry the result. First, the median benefit is positive only in the fast-and-severe corner: at $\tau = 500$ for every multi-peak row (up to +7.3 offline-error units at $h_{\text{sev}} = 3$; sign counts 20–24/30, exact binomial $p < .05$) and at $\tau = 1250$ for $h_{\text{sev}} \geq 3$. Second, everywhere below the boundary the median settles onto a small negative plateau, $-0.02$ to $-0.57$; this plateau *is* the opportunity cost $c(m)$ of the diverted evaluations, read directly from the data ($\hat{c} \approx 0.04$ at the plateau median), exactly as Eq. (1) says the sub-threshold regime must behave. Third, the single-peak anchor row is negative at every $\tau$ (benefit $> 0$ in at most 8/30 seeds, most cells significantly negative): when the static problem gives exploration nothing to find, exploration is overhead at any change rate; the $\Delta$ and $P'$ in Eq. (2) are the quantities that know this.

The sign structure is reproduced by the measured score $S = \hat{\psi}\,\hat{P}'\,\hat{\Delta}$ with a single fitted cutoff shared by all cells: predicted and observed sign agree in 28 of 30 cells (dots in Figure 2), the two misses lying adjacent to the

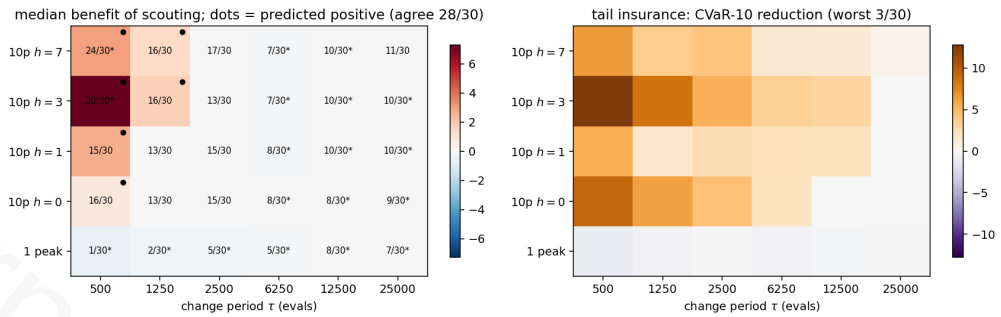



Figure 2: Left: median paired benefit of the scout budget over the change-period × environment plane (30 seeds per cell; annotations: seeds with positive benefit, $*$: exact binomial $p < .05$ in either direction; dots: cells predicted positive by the measured score $\hat{\psi}\hat{P}'\hat{\Delta}$ with one fitted cutoff; agreement 28/30). Right: reduction of CVaR-10 (mean of the worst 3 of 30 runs). The tail benefit extends far below the median boundary and vanishes on the single-peak row.

boundary. On the paired per-seed differences, Wilcoxon signed-rank tests reach $p < .05$ in 16 of 30 cells (the fast-corner positives and the plateau's consistent small negatives; full table in Appendix Appendix A). The claim is not that a new algorithm wins, but that three quantities an optimizer can measure about its own run, combined as Law 1 dictates, locate the boundary between useful exploration and overhead across a two-dimensional family of environments, with one scalar calibrated.

### *4.3. Exploration as insurance*

The mean tells a different story from the median, and the difference is the second result. Over most of the plane the mean benefit is positive while the median is negative; the discrepancy is carried entirely by a heavy upper tail of the error distribution in the $m = 0$ runs: a swarm that converges to the wrong peak and is never rescued accumulates error for the remainder of the run. Scouting removes these runs. Figure 2 (right) quantifies this as the change in CVaR-10, the mean offline error of the worst three runs out of thirty: reductions of +4 to +13 error units persist across the multi-peak rows down to $\tau = 12{,}500$, an order of magnitude beyond the median boundary, and are absent on the single-peak row, where there is nothing to be rescued from. For example at ($\tau$=2500, $h_{\text{sev}}$=3) the median benefit is −0.02 (a pure cost) while the CVaR-10 reduction is +5.5.

The practical reading inverts a habit: in the interior of the phase plane,

a fixed exploration budget should not be defended as improving expected performance (it does not) but as an insurance premium, paid in the median run, that truncates catastrophic stagnation. Where the problem offers no catastrophe (single basin) the premium buys nothing and Law 1 says to cancel the policy.

### *4.4. A marginal-return controller*

Law 1 rearranges into a decision rule: scout while the measured marginal return of a scout evaluation exceeds that of an exploitation evaluation. We implement the comparison with no fitted constants: both returns are exponentially-weighted improvements of the personal-best pool per evaluation, scouts credited with the pool improvement they cause, and a probe floor of one scout per generation keeps the scout estimate alive (one cannot learn the value of exploration without occasionally exploring). The controller switches $m_t$ between the probe floor and the full budget.

Across the same 30-cell plane with matched seeds, median regret against the per-cell better of the two fixed budgets is: fixed $m$=0: +2.44 mean / +8.89 max; triggered hypermutation (50% reinitialise on detection): +0.10 / +3.38; the controller: −0.05 / +3.18 (negative mean: in several mid-plane cells adaptivity beats both fixed budgets); fixed $m$=5: +0.05 / +0.50.

Two conclusions follow. First, on this benchmark the always-on budget is hard to beat, and the reason is in Section 4.3: the premium is tiny ($\hat{c} \approx 0.04$) and the insured catastrophes are not, so the cost–benefit asymmetry favours always-on exploration wherever exploration is cheap, itself a prediction of Eq. (1), not a defeat for it. Second, the controller achieves the oracle-level median while spending a mean budget of $\bar{m} \approx 1.05$ scout evaluations per generation against 5: the same performance at roughly one quarter of the exploration share of the evaluation budget (∼4% versus 16%). When evaluations are expensive, that ratio is the claim; where they are not, Law 1 itself says to stop optimizing the budget and simply pay the premium. The controller's residual weakness is a slow ramp in the fast-and-severe corner (regret up to +3.2 at $\tau \leq 1250$), where volume, not timing, is what the change rate demands; a detection-triggered burst is the obvious refinement and is left for the revision of record.

### *4.5. A standing-exploration architecture under the same accounting*

The reference method for Moving Peaks is a standing-exploration architecture: mQSO [3], here in a light but faithful implementation (10 sub-

swarms of 5+5 quantum particles, exclusion, anti-convergence), run over the same plane with the same seeds. The comparison cuts both ways. On every multi-peak cell, mQSO outperforms our entire single-swarm family, fixed or controlled, with median offline errors of 2.7 to 9.8 against our 8 to 18 (Wilcoxon $p \leq 10^{-3}$ cell-wise): patrolling sub-swarms insure against change and at the same time remove the static deception that a single swarm suffers. On the single-peak anchor the same architecture pays 8.4 to 48.9 error units against the controller's 0.7 to 2.5 ($p = 2 \times 10^{-9}$ in every cell), and the overhead grows as the change period shrinks, because exclusion and anti-convergence keep re-spending evaluations that a single-basin problem cannot repay.

This is Law 1 applied to an architecture rather than a parameter. Standing exploration is the correct purchase where the benefit term is nonzero (multiple basins, at any change rate, since diversity there also repairs the static search) and a recurring cost where it is empty. The accounting does not compete with mQSO; it says where mQSO wins, where it loses, and which quantities such an architecture would need to measure to tell the difference during a run.

### *4.6. The conduit criterion*

Law 3 restricts itself to sampling channels, and both halves of that restriction are empirical results here.

*A negative result, reported as one.* We first tested the variance form on the PSO channel itself: 14,760 change events (channel-only swarms, no local search, no scouts) with the pool variance $V_d$ recorded at each change and recovery judged against the true optimum. Recovery probability depends on displacement as it must (from 25% at shift 1 to 10% at shift 6 on the cleanest subset) but is *flat in* $V_d$: no cliff, no usable $\kappa$. The diagnosis is the scope condition of the law: a constriction swarm's proposals are generated by velocity dynamics, not drawn from the personal-best pool, so pool variance does not bound its reach. We keep the result because it marks the boundary of the claim: the conduit criterion is a property of sampling channels, not of "social" mechanisms in general.

*The sampling case, where the bound is exact.* A recombination channel is a sampling channel in the strict sense: on binary strings, uniform crossover with a removal-only repair can never introduce an item absent from the entire pool, so the value attainable after a change is bounded by the optimum

of the knapsack *restricted to the pool's support*, a quantity computable by dynamic programming from the pool at the instant of change, before any recovery is attempted. We ran 540 single-change experiments (channel-only GA, change injected at $t_c \in \{60, \ldots, 2500\}$ evaluations to vary the pool's convergence state, fixed 2,500-evaluation recovery window). The bound was never violated (540/540). On the 360 events whose pool had fixated (support variance zero, which this greedy steady-state channel reaches within roughly five generations), the bound is attained *exactly*: observed end-of-window loss equals the support-restricted deficit to within 0.01% in 99.2% of events (median discrepancy 0.0000%). In the conduit limit, the channel's future performance is therefore not merely bounded but *determined* by a pool statistic measurable at the change. On the 180 diverse-pool events the bound opens (median reachable deficit 6.2% of the optimum at $t_c$=60) but the channel alone under-exploits it, leaving a median slack of 6.9% between reachable and achieved: diversity makes discovery possible, and closing the remaining gap is the work that Law 1's exploration budget is priced for.

### *4.7. A combinatorial instantiation*

To check that none of this is particular to continuous spaces, we repeat the phase-plane experiment on a dynamic 0/1 knapsack with exact ground truth: $n = 100$ items, integer weights, capacity half the weight sum; every $\tau$ evaluations a fraction $\rho$ of item profits is redrawn, and the moving optimum is recomputed by dynamic programming, so offline error is measured against the true optimum rather than a proxy. The base optimizer is a steady-state GA with greedy repair and a bit-flip local-search budget; the knob is again $m$ random-immigrant scouts per generation; the sentinel re-evaluates the incumbent (with the same generation-start anchoring as before).

Across $\tau \in \{500, 2500, 12500\} \times \rho \in \{0.01, \ldots, 0.5\}$ (30 paired seeds per cell), the gradient structure of Eq. (1) reappears: the median benefit of scouting rises monotonically with both change rate and severity, from +8 profit units (0.2% of the optimum) at the slow, one-item-churn corner to +82 (2.1%) at $\tau$=500, $\rho$=0.5, with the measured $\hat{P}'$ rising fifty-fold along the same diagonal; every cell is positive in 30/30 seeds (exact binomial $p < 10^{-9}$; paired tests in Appendix Appendix A), and the tail (CVaR) reductions again exceed the medians. One structural difference from the continuous domain should be recorded: this GA base is itself statically deceptive (it stalls below the DP optimum even with no changes), so the plane has no negative region; as on the multi-peak continuous rows, exploration here also repairs the static

search. The negative side of the threshold requires a base whose exploitation evaluations have positive marginal value, which is exactly what the cost term $c'$ in Eq. (2) says: $c'$ is the value of the best alternative use of an evaluation, and where the base is stuck that value is not positive. A second domain-specific observation: at $\rho = 0.01$ the sentinel detects only the changes that touch the incumbent's items (∼half), which is the operational meaning of "relevant change rate" in Law 1: changes invisible to the incumbent's value are also the ones that do not drop it.

## 5. Related work

Evolutionary dynamic optimization has accumulated a rich device taxonomy: triggered hypermutation [1], random immigrants [2], explicit memories [9], multi-swarm patrols with exclusion and anti-convergence [3], and prediction-based variants, surveyed at two points a decade apart [6, 7]. Each device embodies a judgement about how much re-exploration the dynamics deserve, and the surveys' recurring complaint is that these judgements are tuned per benchmark rather than derived. The present paper is not another entry in the taxonomy: it is the accounting under which the first two entries (and the exploration component of the others) can be priced from measurements of the run.

Parameter control is the established alternative to hand tuning [5]: self-adaptation encodes parameters in the genome [4], and success-based rules go back to the 1/5th rule [12]. Our controller belongs to this family but differs in what it equalizes: not a success *rate* against a fixed target, but the measured marginal return of two uses of an evaluation, which is Law 1 rearranged. The 1/5th rule survives inside the construction as the estimator of $P'$.

Multi-swarm methods such as mQSO [3] maintain standing exploration (patrolling sub-swarms and anti-convergence reinjection) regardless of the measured dynamics. In the vocabulary of Section 4.3, they pay the insurance premium permanently. Our results say when that is rational (cheap evaluations, multi-basin problems, any nonzero change rate) and when it is overhead (single-basin or static regimes), and provide the quantities that distinguish the cases online.

The budget split between population search and local refinement is the central design axis of memetic algorithms [13]. Our prototype sequence forced

a distinction that literature often leaves implicit: refinement around the incumbent is exploitation and carries no threshold; the threshold governs *global* exploration only. Finally, risk-sensitive evaluation via conditional value-at-risk [8] is standard in finance but rare as a reporting practice in evolutionary computation; Section 4.3 suggests it is the correct unit of account for exploration over most of the phase plane.

## 6. Discussion

What do the two laws buy that the device taxonomy does not? A sign prediction, checked cell by cell: three online-measurable quantities reproduce the boundary of useful exploration in 28/30 environments of the continuous plane with one calibrated scalar, reproduce its monotone structure on a combinatorial domain with exact ground truth, and, for sampling channels, determine the post-change performance of the sharing machinery exactly in the converged limit. None of these statements names an algorithm; they constrain every algorithm built from these parts.

Two empirical complications are findings rather than nuisances. First, "exploration helps" conflates two services: insuring against environmental change and repairing a statically deceptive search. The cost term $c'$ separates them: it is the marginal value of the best alternative use of an evaluation, and where the base optimizer is stuck that value is not positive, so the threshold collapses and exploration is always justified. A practitioner crediting a dynamic mechanism should first measure the base's static health; our single-peak and single-basin anchors are the template. Second, the mean is the wrong summary over most of the phase plane: exploration's value there is a truncation of the catastrophic tail, invisible to expected offline error and natural in CVaR. We suggest dynamic-optimization studies report both.

The conduit results say what "diversity maintenance" is for. A recombination channel's reachable set is its pool's span, and in the converged limit its future loss is computable at the instant of change. Diversity does not itself find anything (the diverse-pool slack of 6.9% measures that) but it is the precondition for the exploration budget to have anything to recombine with. The PSO negative result bounds the claim: mechanisms with their own proposal dynamics are not covered, and deciding which channels in the swarm literature are sampling channels (estimation-of-distribution models and pheromone constructions are) is an audit we leave open.

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*Limitations.* The phase planes use one benchmark family per domain and a binary budget ($m \in \{1, 5\}$); the controller is minimal by construction and ramps slowly where volume matters; the sentinel measures *relevant* change by construction, which is the quantity the law wants but will under-estimate $\psi$ for changes invisible to the incumbent until they bind. All code, archived outputs and figure scripts are public at https://github.com/drnamlabs-gif/two-laws.

## Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used AI-assisted tools to paraphrase text, check grammar and rewrite passages. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

## Appendix A. Statistical tables

All comparisons are paired by seed: the two members of a pair share the same initial population and the same change schedule, so the paired $t$ and Wilcoxon signed-rank tests on per-seed differences are the appropriate instruments, and $d_z$ is the paired effect size (mean difference over the standard deviation of differences).

Table A.1: Moving Peaks: absolute offline error, no exploration ($m$=0) vs always-on ($m$=5), 30 seeds per cell. Paired $t$ and Wilcoxon signed-rank $p$ (two-sided), paired effect size $d_z$ ($d > 0$: exploration better).

| $\tau$ | env | $m$=0 | $m$=5 | $p_t$ | $p_{Wx}$ | $d$ |
|---|---|---|---|---|---|---|
| 500 | 1 peak | $2.45 \pm 0.77$ | $3.02 \pm 0.97$ | 1.04e-11*** | 3.73e-09*** | -1.98 |
| 500 | 10p $h$=0 | $16.35 \pm 12.66$ | $10.08 \pm 8.07$ | 0.000141*** | 0.0175* | +0.80 |
| 500 | 10p $h$=1 | $16.90 \pm 11.80$ | $11.08 \pm 7.69$ | 6.67e-05*** | 0.0197* | +0.85 |
| 500 | 10p $h$=3 | $18.91 \pm 9.13$ | $12.86 \pm 4.92$ | 1.85e-06*** | 9.9e-05*** | +1.09 |
| 500 | 10p $h$=7 | $19.21 \pm 5.40$ | $15.34 \pm 4.27$ | 0.000292*** | 4.41e-05*** | +0.75 |
| 1250 | 1 peak | $1.98 \pm 0.80$ | $2.22 \pm 0.94$ | 1.81e-06*** | 8.33e-07*** | -1.09 |
| 1250 | 10p $h$=0 | $15.74 \pm 12.69$ | $11.02 \pm 9.21$ | 0.00103** | 0.328 | +0.67 |
| 1250 | 10p $h$=1 | $16.56 \pm 12.89$ | $11.83 \pm 10.22$ | 0.00239** | 0.299 | +0.61 |

Table A.1: Moving Peaks: absolute offline error, no exploration ($m$=0) vs always-on ($m$=5), 30 seeds per cell. Paired $t$ and Wilcoxon signed-rank $p$ (two-sided), paired effect size $d_z$ ($d > 0$: exploration better).

| $\tau$ | env | $m$=0 | $m$=5 | $p_t$ | $p_{Wx}$ | $d$ |
|---|---|---|---|---|---|---|
| 1250 | 10p $h$=3 | $18.04 \pm 11.63$ | $12.67 \pm 7.90$ | 0.000139*** | 0.0175* | +0.80 |
| 1250 | 10p $h$=7 | $19.45 \pm 8.91$ | $15.92 \pm 7.24$ | 0.000253*** | 0.00871** | +0.76 |
| 2500 | 1 peak | $1.67 \pm 0.73$ | $1.86 \pm 0.80$ | 0.000707*** | 0.000555*** | -0.69 |
| 2500 | 10p $h$=0 | $15.44 \pm 12.71$ | $12.82 \pm 10.58$ | 0.00188** | 0.119 | +0.62 |
| 2500 | 10p $h$=1 | $16.43 \pm 13.05$ | $13.42 \pm 10.73$ | 0.00387** | 0.119 | +0.57 |
| 2500 | 10p $h$=3 | $18.30 \pm 12.34$ | $15.37 \pm 9.80$ | 0.0083** | 0.328 | +0.52 |
| 2500 | 10p $h$=7 | $20.35 \pm 10.57$ | $16.68 \pm 8.35$ | 0.00212** | 0.0667 | +0.62 |
| 6250 | 1 peak | $1.01 \pm 0.42$ | $1.19 \pm 0.53$ | 5.05e-05*** | 1.22e-05*** | -0.87 |
| 6250 | 10p $h$=0 | $15.03 \pm 12.69$ | $14.25 \pm 11.97$ | 0.111 | 0.0767 | +0.30 |
| 6250 | 10p $h$=1 | $15.61 \pm 12.73$ | $14.63 \pm 11.83$ | 0.0678 | 0.0577 | +0.35 |
| 6250 | 10p $h$=3 | $16.74 \pm 12.27$ | $15.35 \pm 11.12$ | 0.0671 | 0.0327* | +0.35 |
| 6250 | 10p $h$=7 | $18.72 \pm 11.28$ | $16.92 \pm 10.36$ | 0.09 | 0.0197* | +0.32 |
| 12500 | 1 peak | $0.84 \pm 0.41$ | $0.95 \pm 0.46$ | 0.0325* | 0.0106* | -0.41 |
| 12500 | 10p $h$=0 | $14.94 \pm 12.67$ | $14.79 \pm 12.56$ | 0.419 | 0.00871** | +0.15 |
| 12500 | 10p $h$=1 | $15.38 \pm 12.76$ | $14.68 \pm 12.03$ | 0.102 | 0.105 | +0.31 |
| 12500 | 10p $h$=3 | $16.07 \pm 12.77$ | $15.07 \pm 11.72$ | 0.0546 | 0.245 | +0.37 |
| 12500 | 10p $h$=7 | $17.19 \pm 12.42$ | $16.09 \pm 11.63$ | 0.153 | 0.135 | +0.27 |
| 25000 | 1 peak | $0.75 \pm 0.38$ | $0.79 \pm 0.40$ | 0.312 | 0.012* | -0.19 |
| 25000 | 10p $h$=0 | $14.91 \pm 12.67$ | $14.93 \pm 12.67$ | 0.0141* | 0.0185* | -0.48 |
| 25000 | 10p $h$=1 | $15.13 \pm 12.77$ | $14.92 \pm 12.58$ | 0.374 | 0.0699 | +0.16 |
| 25000 | 10p $h$=3 | $15.64 \pm 12.93$ | $15.37 \pm 12.70$ | 0.364 | 0.0549 | +0.17 |
| 25000 | 10p $h$=7 | $16.41 \pm 13.01$ | $15.85 \pm 12.46$ | 0.206 | 0.191 | +0.24 |

Table A.2: Moving Peaks: mean offline error of the four policies (30 seeds), and Wilcoxon signed-rank $p$ of controller vs each alternative.

| $\tau$ | env | $m$=0 | $m$=5 | hyper | ctrl | $p$ vs $m$0 | vs $m$5 | vs hyp |
|---|---|---|---|---|---|---|---|---|
| 500 | 1 peak | 2.5 | 3.0 | 3.0 | 2.5 | 0.088 | 1.9e-09*** | 1.9e-09*** |
| 500 | 10p $h$=0 | 16.4 | 10.1 | 10.5 | 12.9 | 0.37 | 0.28 | 0.87 |

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Table A.2: Moving Peaks: mean offline error of the four policies (30 seeds), and Wilcoxon signed-rank $p$ of controller vs each alternative.

| $\tau$ | env | $m$=0 | $m$=5 | hyper | ctrl | $p$ vs $m$0 | vs $m$5 | vs hyp |
|---|---|---|---|---|---|---|---|---|
| 500 | 10p $h$=1 | 16.9 | 11.1 | 10.9 | 14.2 | 0.75 | 0.58 | 0.16 |
| 500 | 10p $h$=3 | 18.9 | 12.9 | 13.3 | 14.9 | 0.0076** | 0.12 | 0.073 |
| 500 | 10p $h$=7 | 19.2 | 15.3 | 15.6 | 16.5 | 2.4e-05*** | 0.031* | 0.12 |
| 1250 | 1 peak | 2.0 | 2.2 | 1.9 | 1.9 | 0.61 | 2.6e-07*** | 0.025* |
| 1250 | 10p $h$=0 | 15.7 | 11.0 | 11.5 | 12.4 | 0.11 | 0.56 | 0.00038*** |
| 1250 | 10p $h$=1 | 16.6 | 11.8 | 11.8 | 13.2 | 0.058 | 0.97 | 0.0058** |
| 1250 | 10p $h$=3 | 18.0 | 12.7 | 14.0 | 13.3 | 0.013* | 0.9 | 0.63 |
| 1250 | 10p $h$=7 | 19.5 | 15.9 | 16.7 | 16.3 | 0.0015** | 0.86 | 0.6 |
| 2500 | 1 peak | 1.7 | 1.9 | 1.6 | 1.7 | 0.38 | 6.3e-05*** | 0.0032** |
| 2500 | 10p $h$=0 | 15.4 | 12.8 | 13.4 | 13.4 | 0.54 | 0.53 | 0.038* |
| 2500 | 10p $h$=1 | 16.4 | 13.4 | 13.6 | 13.8 | 0.32 | 0.6 | 0.21 |
| 2500 | 10p $h$=3 | 18.3 | 15.4 | 15.4 | 15.1 | 0.08 | 0.16 | 0.31 |
| 2500 | 10p $h$=7 | 20.3 | 16.7 | 17.0 | 16.7 | 0.11 | 0.63 | 0.061 |
| 6250 | 1 peak | 1.0 | 1.2 | 1.2 | 1.1 | 0.11 | 0.00021*** | 0.17 |
| 6250 | 10p $h$=0 | 15.0 | 14.2 | 14.1 | 14.0 | 0.077 | 0.23 | 0.82 |
| 6250 | 10p $h$=1 | 15.6 | 14.6 | 14.5 | 14.2 | 0.16 | 0.18 | 0.39 |
| 6250 | 10p $h$=3 | 16.7 | 15.3 | 15.5 | 15.1 | 0.092 | 0.22 | 0.46 |
| 6250 | 10p $h$=7 | 18.7 | 16.9 | 17.9 | 16.9 | 0.31 | 0.12 | 0.23 |
| 12500 | 1 peak | 0.8 | 1.0 | 0.9 | 0.9 | 0.33 | 0.016* | 0.69 |
| 12500 | 10p $h$=0 | 14.9 | 14.8 | 14.8 | 13.9 | 0.66 | 0.15 | 0.66 |
| 12500 | 10p $h$=1 | 15.4 | 14.7 | 15.2 | 14.3 | 0.66 | 0.32 | 0.7 |
| 12500 | 10p $h$=3 | 16.1 | 15.1 | 15.9 | 14.8 | 0.66 | 0.57 | 0.67 |
| 12500 | 10p $h$=7 | 17.2 | 16.1 | 16.5 | 15.7 | 0.75 | 0.3 | 0.52 |
| 25000 | 1 peak | 0.7 | 0.8 | 0.7 | 0.8 | 0.61 | 0.061 | 0.48 |
| 25000 | 10p $h$=0 | 14.9 | 14.9 | 14.9 | 13.9 | 0.9 | 0.016* | 0.87 |
| 25000 | 10p $h$=1 | 15.1 | 14.9 | 15.1 | 14.1 | 0.92 | 0.016* | 0.84 |
| 25000 | 10p $h$=3 | 15.6 | 15.4 | 15.6 | 14.5 | 0.86 | 0.073 | 0.84 |
| 25000 | 10p $h$=7 | 16.4 | 15.9 | 16.2 | 15.1 | 0.89 | 0.07 | 0.4 |

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Table A.3: Dynamic knapsack: mean offline error (and % of the DP optimum), $m$=0 vs $m$=5, 30 seeds. Paired $t$ and Mann–Whitney $p$, Cohen's $d$.

| $\tau$ | $\rho$ | $m$=0 | $m$=5 | $p_t$ | $p_{Wx}$ | $d$ |
|---|---|---|---|---|---|---|
| 500 | 0.01 | 28.0 (0.70%) | 18.8 (0.47%) | 4.1e-13*** | 1.9e-09*** | +2.26 |
| 500 | 0.05 | 34.1 (0.85%) | 22.6 (0.56%) | 5e-20*** | 1.9e-09*** | +4.15 |
| 500 | 0.2 | 78.7 (1.93%) | 50.9 (1.25%) | 2.1e-32*** | 1.9e-09*** | +11.34 |
| 500 | 0.5 | 179.6 (4.38%) | 97.9 (2.39%) | 8.3e-33*** | 1.9e-09*** | +11.71 |
| 2500 | 0.01 | 22.7 (0.56%) | 11.6 (0.29%) | 3.1e-12*** | 1.9e-09*** | +2.08 |
| 2500 | 0.05 | 20.9 (0.51%) | 10.5 (0.26%) | 2.4e-18*** | 1.9e-09*** | +3.60 |
| 2500 | 0.2 | 30.8 (0.76%) | 15.5 (0.38%) | 6.3e-22*** | 1.9e-09*** | +4.86 |
| 2500 | 0.5 | 51.7 (1.27%) | 24.7 (0.61%) | 1.4e-27*** | 1.9e-09*** | +7.70 |
| 12500 | 0.01 | 17.2 (0.43%) | 5.6 (0.14%) | 5.7e-11*** | 1.9e-09*** | +1.84 |
| 12500 | 0.05 | 16.9 (0.42%) | 5.8 (0.14%) | 2.2e-13*** | 1.9e-09*** | +2.32 |
| 12500 | 0.2 | 19.4 (0.49%) | 7.4 (0.19%) | 1.6e-10*** | 1.9e-09*** | +1.76 |
| 12500 | 0.5 | 23.3 (0.58%) | 8.6 (0.21%) | 4.5e-14*** | 1.9e-09*** | +2.47 |

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